The Return to Schooling for Married Women, 1975

ECON 3916 Midterm Project · Nov 8, 2026 ·David Xu

1. The Question and the Data

Question. Among married women in the U.S. who worked for pay in 1975, how much higher were the hourly wage associated with one year of schooling, experience, the number of children under 6, and mother's schooling constant? The outcome is the log of the hourly wage so the schooling coefficient is to be bead as an approximate estimate in percentage difference.

Why it matters. The return to schooling is the price the labor market puts on education; it matters to women deciding whether school will pay off or not, to policymakers funding education and student aid, and to economists studying the gender wage gap, since most early estimates came from samples of men.Starting in the mid-1970s is a useful set of baseline as married woman were the main group who associated with paid work were changing the fastest.

The data are a product of measurement decisions. These data are from the Panel Study of Income Dynamics, a University of Michigan household survey in 1968 to study poverty. Mroz drew 753 married white women from ages 30-60 observed in 1795 to study how many hours they work. Woolbridge distributes the file for teaching. Four decisions shape what they mean:

- *The wage was never asked directly.* It is annual earnings divided by annual hours, so any error in reported hours becomes error in the wage.
- *Schooling is years completed,* with no measure of degrees or school quality.
- *Experience is actual years worked,* recalled by the respondent. That suits women with career breaks but carries recall error.
- *The sample is only married white women aged 30–60,* and dollars are nominal 1975 dollars.

What is missing, and why? Every variable is complete except the wage and its log. These are missing for the 325 women (43%) who did not work for pay: with no earnings, there is nothing to divide by hours. The regressions use the 428 women who worked.

The mechanism is MNAR. The wages are not missing completely at random; women with a wage have .86 more years of schooling, 5.58 more years of experience, 0.226 fewer children

under 6, and $2.76 thousand less other family income than women without one, all significant at the 1% level. Nor is it merely missing at random, because missingness depends on the wage itself. A woman takes a job when the offer is worth more than her time spent at home, so the women who do not work are disproportionately those offered lower wages. That is missing not at random. This can't be proven since the missing wages are never observed, but it is the standard.

What this does to the answer. The estimate describes working married women. Less-educated women work only when their offers are unusually good, which pulls up their observed wages and flattens the slope, so the estimate is probably a lower bound for all married women. I do not impute the missing wages, because standard imputation assumes MAR.

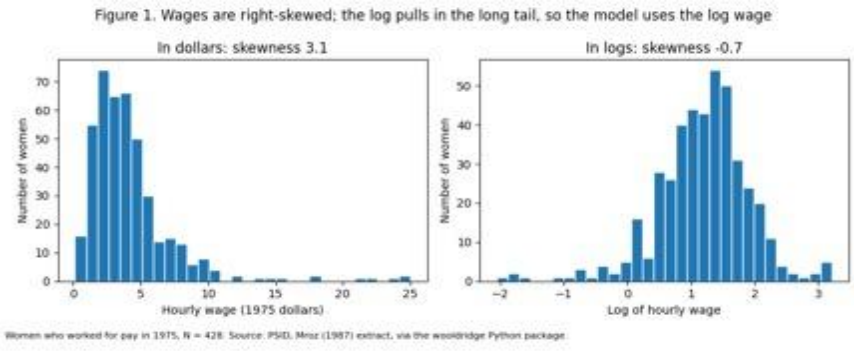
Table 1. Summary statistics, 753 married white women aged 30–60, PSID, 1975

Variable	N	Mean	SD	Min	Median	Max
Hourly wage (1975 $)	428	4.18	3.31	0.13	3.48	25.00
Log hourly wage	428	1.19	0.72	−2.05	1.25	3.22
Schooling (years completed)	753	12.29	2.28	5	12	17
Actual labor-market experience (years)	753	10.63	8.07	0	9	45
Children under 6 (number)	753	0.24	0.52	0	0	3
Mother's schooling (years)	753	9.25	3.37	0	10	17
Age (years)	753	42.54	8.07	30	43	60
Family income excluding her earnings ($1,000s)	753	20.13	11.63	−0.03	17.70	96.00
Worked for pay in 1975 (share)	753	0.57	0.50	0	1	1

2. Descriptive Evidence

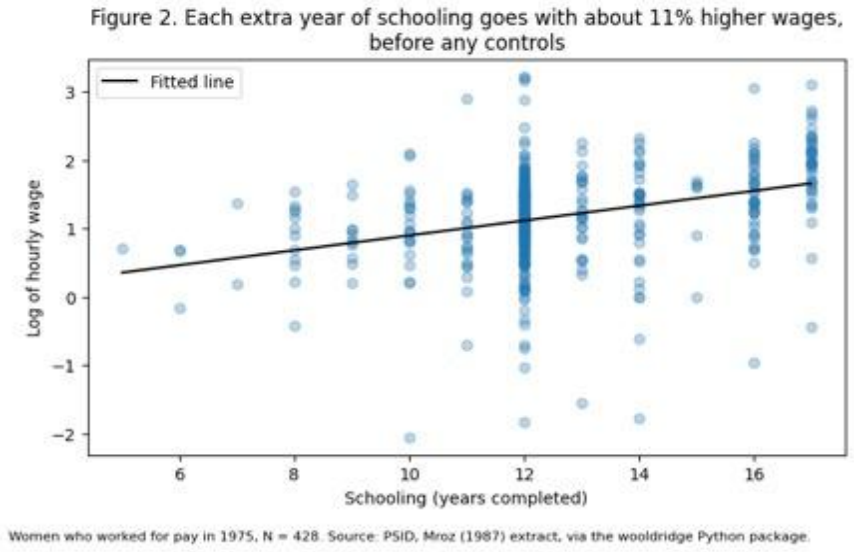
Three patterns shape the model: wages are skewed, they rise with schooling, and who has a wage at all depends on family circumstances.

Figure 1. Wages are right-skewed; the log pulls in the long tail, so the model uses the log wage.



The hourly wage has a long right tail, reaching $25, and a skewness of 3.09; in logs, the skewness is −0.69. By modeling the log wages, it makes the coefficients percent differences, the natural scale for a return to schooling. The log exposes a short left tail of low wages, most likely due to errors in reported hours.

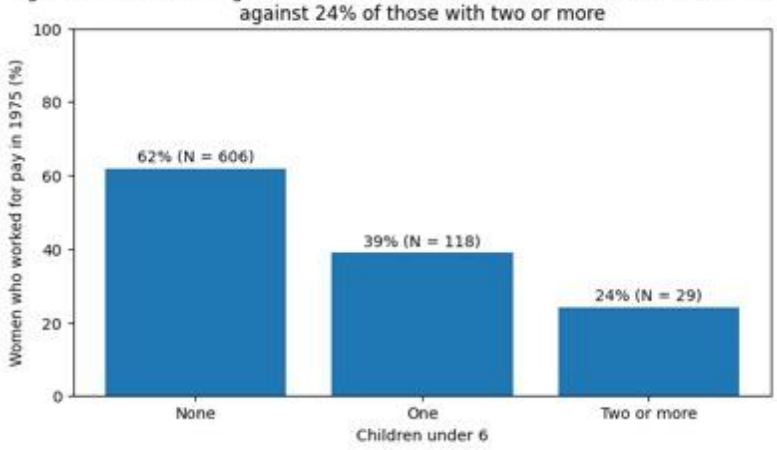
Figure 2. Each extra year of schooling goes with about 11.5% higher wages, before any controls.



In the raw data, one more year of schooling goes with an 11.5% higher hourly wage. Most women have 12 years of school, so the middle of the range pins down the line. Section 3 asks how much of this association survives the controls.

Figure 3. Who has a wage is not random: 62% of women with no child under 6 worked, against 24% of those with two or more.

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All 753 women in the sample. Source: PSID, Mroz (1987) extract, via the wooldridge Python package.

Women with young children were far less likely to work for pay, so the 428 observed wages come from a selected group: the selection described in Section 1, seen directly. All three figures label axes with units, start the bar chart at zero, state the sample and source, and put the finding in the title.

3. The Model

Coefficient 1: schooling. Based upon a mother's schooling, experience, and young children constant, one more year of schooling results in a 12.6% higher hourly wage. The coefficient is 0.1189 long points. The robust 95% confidcens runs an interval from 9.4% to 16%, and even its low end its economically large. At the average wage of $4.18, that is about 53 cents more per hour in 1975.

Coefficient 2: children under 6. Each additional child under 6 means the result of a 5.4% lower hourly wage, holding the rest constant, with a coefficient of -.0560 and a robust SE of 0.0981. The 95% interval runs between 22% lower and 14.6% higher, so the data can't rule out no association. Other factors come into play, such as that free working women had young children, which is why the estimate is not very precise.

Experience and fit. Experience raises wages at a falling rate, with the curve peaking at about 24.7 years. The model explains 16.4% of the variation in log wages (R-squared 0.164), typical for wage regressions.

The role of each control.

- *Mother's schooling* blocks a confounding path. Family background shapes both a woman's schooling and, through expectations and contacts, her earnings, and mother's schooling is correlated 0.39 with her own. Her schooling cannot change her mother's, so it cannot be a mediator; its weakness is that it captures background only partly and ability not at all.

- *Experience and its square* mainly add precision, because experience strongly predicts wages. They would be a bad control if schooling raises wages partly by raising experience, since controlling for experience removes that channel.
- *Children under 6* block a confounding path, because family and career preferences shape both schooling and job choice. It would be a bad control if schooling changes fertility, for example by leading women to delay having children.

What the controls did. Adding mother's schooling raised the schooling coefficient from 0.1086 to 0.1210, about a 1.2% increase. Holding her own schooling fixed, mother's schooling is slightly negative based on wages being −0.0184, p = 0.095, so omitting it biased the schooling coefficient down. One reading is that women from more educated families had more household resources and could select jobs for reasons beyond the hourly pay, but at p = .095 it may also be a distraction. Adding experience and children under 6 doesn't move the coefficient much from 0.1176 to 0.1189, so the possible mediators don't change the answer.

Left out on purpose. Hours worked is part of how the wage is built. Husband's schooling and wage are partly consequences of the woman's schooling, through whom she marries. Family income contains her own earnings. Each would be a bad control.

4. Diagnostics

Residuals. Plotted against the fitted values and against schooling, the residuals are centered on zero with no clear curve, which makes the linear form reasonable. A few large negative residuals come from the implausibly low wages. The residuals' skewness of −0.73 and kurtosis of 6.7 give a much heavier lower tail, which matters very little with this large number of 428 observations.

Constant variance. The Breusch-Pagan statistic is 13.51 with p = .019, so constant variance stands at the 5% level. Every standard error in Table 3 is therefore heteroskedasticity-robust. Robust and conventional errors for schooling are relatively the same, which means the main result does not depend on this choice.

Multicollinearity. Schooling's VIF is 1.20, far below the usual thresholds of 5 or 10. Experience and its square have VIFs of about 10.9 only because one is the square of the other. They are interpreted together as one curve, so this is harmless.

Influence. Twenty-four women exceed the Cook's distance cutoff of 4/N (Figure 5, Appendix), and the most influential earned far under a dollar an hour. Dropping all 24 moves the schooling coefficient from 0.1189 to 0.1173, well inside its confidence interval, so they do not drive the result. I keep them as genuine survey responses. Together, these checks support the estimate as a description of this sample; they cannot address omitted ability or selection into work, which Section 5 takes up.

5. Threats and Conclusion

Threats to a causal reading. The schooling coefficient would be causal only if, among women with the same mother's schooling, experience, and young children, schooling were unrelated to earning power. That fails for three reasons.

1. **Omitted ability.** More able or motivated women stay in school longer and would earn more anyway. This pushes the estimate *up*.
2. **Selection into work.** Wages are missing not at random, and less-educated women who work tend to have unusually good offers. This flattens the slope and pushes the estimate *down* relative to all married women.
3. **Measurement.** Self-reported schooling ignores school quality, and random error in it pulls the estimate toward zero.

The biases pull in opposite directions, so the net bias cannot be signed.

What the estimate is instead. It is a **conditional association**: the average difference in log hourly wages between married white women who worked in 1975 and differ by one year of schooling but share mother's schooling, experience, and number of young children. It is descriptive, not causal, and says nothing about women who did not work.

Conclusion. Among these women, one more year of schooling is associated with about a 12.6% higher hourly wage (95% CI 9.4% to 16.0%). It rose about 1.2 percentage points when mother's schooling was added, barely moved with the other controls, and stayed inside its confidence interval when the influential women were dropped. As a description of this sample, it is solid; as a causal effect, it deserves much less confidence.

With another month, I would estimate a Heckman selection model, using young children and other family income to shift participation, look for variation in schooling unrelated to ability, such as distance to college, and repeat the analysis on later PSID years and broader samples. Parents' schooling would be a doubtful instrument, since mothers' schooling is also associated with wages.

Appendix

The appendix holds the supporting tables and the diagnostic figures; it does not count toward the report's page limit.

Table 2. Means for women whose wage is observed and missing

Variable	Wage observed (worked, N = 428)	Wage missing (did not work, N = 325)	Difference	p-value
Schooling (years)	12.66	11.80	+0.86	0.000

Experience (years)	13.04	7.46	+5.58	0.000
Age (years)	41.97	43.28	−1.31	0.029
Children under 6	0.14	0.37	−0.23	0.000
Children aged 6–18	1.35	1.36	−0.01	0.947
Mother's schooling (years)	9.52	8.90	+0.62	0.013
Other family income ($1,000s)	18.94	21.70	−2.76	0.002
Husband's schooling (years)	12.61	12.33	+0.28	0.207
Husband's hourly wage (1975 $)	7.23	7.82	−0.59	0.068

Notes: Difference = observed minus missing. p-values from two-sided Welch t-tests. Under MCAR the two groups would look alike on every row.

Table 3. OLS estimates. Dependent variable: log hourly wage (1975 dollars)

	(1)	(2)	(3)	(4) Main
Schooling (years)	0.1086***	0.1210***	0.1176***	0.1189***
	(0.0134)	(0.0148)	(0.0146)	(0.0149)
Mother's schooling (years)		−0.0220**	−0.0183*	−0.0184*
		(0.0110)	(0.0109)	(0.0110)
Experience (years)			0.0415***	0.0408***
			(0.0151)	(0.0150)
Experience squared			−0.0008**	−0.0008**
			(0.0004)	(0.0004)
Children under 6 (number)				−0.0560
				(0.0981)
Constant	−0.1852	−0.1317	−0.4702**	−0.4699**
	(0.1707)	(0.1732)	(0.2044)	(0.2043)
Observations	428	428	428	428
R-squared	0.118	0.127	0.163	0.164

Notes: Married white women who worked for pay in 1975 (PSID, Mroz 1987 extract). Robust (HC1) standard errors in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.10.

Figure 4. Residuals against fitted values and against schooling.

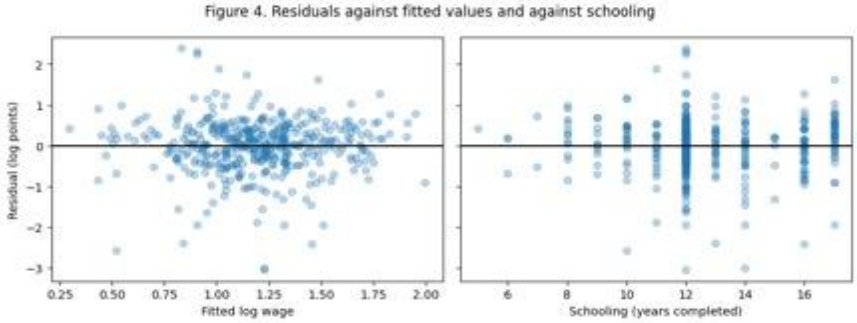
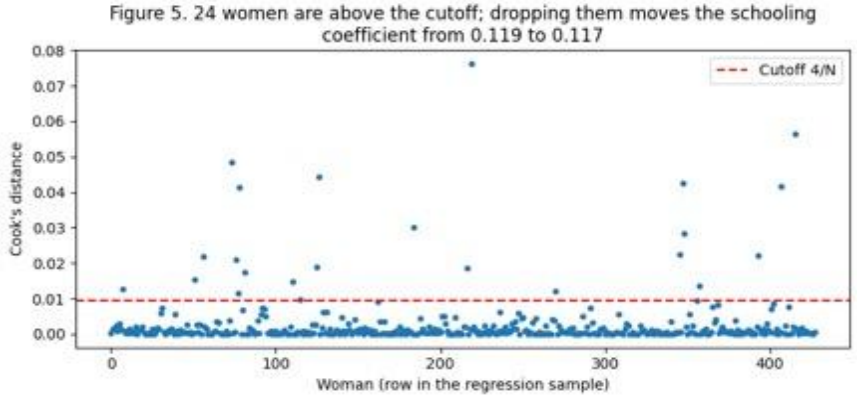


Figure 5. 24 women are above the Cook's distance cutoff; dropping them moves the schooling coefficient from 0.119 to 0.117.



References and Reproducibility

- Heckman, J. (1979). Sample selection bias as a specification error. *Econometrica*, 47(1), 153–161.
- Mroz, T. (1987). The sensitivity of an empirical model of married women's hours of work to economic and statistical assumptions. *Econometrica*, 55(4), 765–799.
- Data: Mroz dataset from the wooldridge Python package, downloaded 2026-10-26; original source: the Panel Study of Income Dynamics, University of Michigan.